LABORATORY OBSERVATION / RECORD

Course: Full Stack Web Development
Course Code: 23CM4121
Branch: CSE-AIML
Experiment No.: 2
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1. TITLE

Design Webpage using HTML5 & CSS3

2. AIM

To design and develop a responsive webpage incorporating HTML5 graphics (Canvas API) and advanced CSS3 styling techniques including gradients, box shadows, border-radius, transitions, and hover effects.

3. OBJECTIVE

- To understand dynamic graphic rendering using HTML5 <canvas> and JavaScript 2D Context API.
- To implement modern CSS3 styling properties including CSS Grid/Flexbox layouts, gradients, box-shadow, and custom typography.
- To build interactive components with CSS3 transition and transform hover effects.
- To structure semantic web content alongside visual enhancement cards and customized navigation bars.

4. THEORY

HTML5 Canvas: The HTML5 <canvas> element provides an executable region for drawing dynamic graphics via JavaScript scripting. Using the 2D rendering context (getContext('2d')), developer APIs allow programmatically drawing geometric paths, shapes (rectangles, arcs/circles, lines), filling colors, defining strokes, and rendering custom typographic text directly onto the visual DOM layer.

CSS3 Styling and Enhancements: Cascading Style Sheets Level 3 (CSS3) provides sophisticated user interface design capabilities without relying on external image assets. Key features implemented include:
• Linear Gradients: Blending multiple color stops seamlessly for backgrounds.

• Box & Text Shadows: Adding depth and elevation effects (box-shadow, text-shadow).
• Border Radius: Rounding element borders for modern UI design aesthetic.
• Transitions & Transforms: Animating element states smoothly on user interactions like hover (translateY, scale).
• Flexbox Container: Laying out component cards dynamically with flexible spacing and alignment rules.

5. ALGORITHM/PROCEDURE

1. Create a project workspace with index.html for structural elements and style.css for external visual styling rules.

2. Structure index.html with standard semantic wrappers: <header>, <nav>, <section>, and <footer>.

3. In the header section, introduce page titles with CSS linear gradient and text-shadow accents.

4. Build a navigation bar (<nav>) housing horizontal navigation hyperlinks styled with rounded border radius and hover state color shifts.

5. Include an HTML5 <canvas id="myCanvas"> section with predefined canvas dimensions (500x300).

6. Implement inline JavaScript script tags to draw shapes on the canvas context: a styled blue rectangle, a yellow circle, a baseline stroke line, and text displaying author identity.

7. Construct a Content Cards container utilizing CSS Flexbox to position three distinct card elements side-by-side with border-radius, linear-gradients, and custom button triggers.

8. Apply CSS hover transitions (`transform: translateY(-8px)` and `scale(1.08)`) for responsive interaction feedback.

9. Attach a <footer> displaying author copyright information.

10. Load index.html in a modern browser to verify Canvas rendering and CSS interactive behaviors.

6. PROGRAM

index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>My Styled Webpage</title>
```

```html
    <link rel="stylesheet" href="./style.css">
  </head>
  <body>
    <!-- ===== Header ===== -->
    <header>
      <h1>My Styled Webpage</h1>
      <p>HTML5 Graphics and CSS3 Web Styling</p>
    </header>

    <!-- ===== Navigation Bar ===== -->
    <nav>
      <a href="#home">Home</a>
      <a href="#about">About</a>
      <a href="#services">Services</a>
      <a href="#contact">Contact</a>
    </nav>

    <!-- ===== Part A: HTML5 Canvas ===== -->
    <section class="canvas-section">
      <h2>Part A: HTML5 Canvas Drawing</h2>
      <canvas id="myCanvas" width="500" height="300"></canvas>
    </section>

    <!-- ===== Part B: Content Cards ===== -->
    <section class="cards-container">
      <div class="card">
        <h2>Card One</h2>
        <p>This card demonstrates border radius, box shadow, and hover effects using CSS3.</p>
        <button>Learn More</button>
      </div>
      <div class="card">
        <h2>Card Two</h2>
        <p>Gradient backgrounds and text shadows are applied to enhance visual appeal.</p>
        <button>Learn More</button>
      </div>
      <div class="card">
        <h2>Card Three</h2>
        <p>Margins, padding, and fonts are combined to create a clean card layout.</p>
        <button>Learn More</button>
      </div>
    </section>

    <!-- ===== Footer ===== -->
    <footer>
```

```
        &copy; 2026 My Styled Webpage | Created by Vijay Anudeep
    </footer>

    <!-- ===== JavaScript for Canvas Drawing ===== -->
    <script>
        const canvas = document.getElementById('myCanvas');
        const ctx = canvas.getContext('2d');

        // 1. Draw a Rectangle
        ctx.fillStyle = '#4b6cb7';
        ctx.fillRect(30, 30, 150, 80);
        ctx.strokeStyle = '#182848';
        ctx.lineWidth = 3;
        ctx.strokeRect(30, 30, 150, 80);

        // 2. Draw a Circle
        ctx.beginPath();
        ctx.arc(300, 70, 50, 0, Math.PI * 2);
        ctx.fillStyle = '#ffe600';
        ctx.fill();
        ctx.strokeStyle = '#182848';
        ctx.lineWidth = 3;
        ctx.stroke();

        // 3. Draw a Line
        ctx.beginPath();
        ctx.moveTo(30, 180);
        ctx.lineTo(450, 180);
        ctx.strokeStyle = '#182848';
        ctx.lineWidth = 4;
        ctx.stroke();

        // 4. Display Name as Text
        ctx.font = "bold 30px 'Georgia', serif";
        ctx.fillStyle = '#182848';
        ctx.fillText("Vijay Anudeep", 130, 250);
    </script>
</body>
</html>
```

style.css

```
/* ===== CSS Syntax / Selectors / Colors / Fonts ===== */
* {
    box-sizing: border-box;
```

```css
}

body {
    margin: 0;
    padding: 0;
    font-family: 'Segoe UI', Arial, sans-serif;
    background-color: #f0f2f5;
    color: #222222;
}

/* ===== Header ===== */
header {
    background: linear-gradient(to right, #4b6cb7, #182848); /* Gradient background */
    color: #ffffff;
    text-align: center;
    padding: 30px 20px;
    text-shadow: 2px 2px 4px rgba(0, 0, 0, 0.6); /* Text Shadow */
}

header h1 {
    font-family: 'Georgia', serif;
    font-size: 40px;
    margin: 0;
}

header p {
    font-family: 'Verdana', sans-serif;
    font-size: 16px;
    margin-top: 10px;
}

/* ===== Navigation Bar ===== */
nav {
    background-color: #1f1f2e;
    padding: 15px 0;
    text-align: center;
    box-shadow: 0px 4px 8px rgba(0, 0, 0, 0.3); /* Box Shadow */
}

nav a {
    color: #ffffff;
    text-decoration: none;
    font-family: 'Trebuchet MS', sans-serif;
    font-size: 18px;
```

```
    margin: 0 20px;
    padding: 8px 16px;
    border-radius: 6px; /* Rounded Corners */
    transition: all 0.3s ease-in-out;
}

/* Hover effect for navigation links */
nav a:hover {
    background-color: #4b6cb7;
    color: #ffe600;
    text-shadow: 1px 1px 3px #000000;
}

/* ===== Canvas Section ===== */
.canvas-section {
    text-align: center;
    margin: 40px auto;
    padding: 20px;
}

canvas {
    background-color: #ffffff;
    border: 3px solid #4b6cb7; /* Border */
    border-radius: 12px; /* Rounded Corners */
    box-shadow: 4px 4px 12px rgba(0, 0, 0, 0.25); /* Box Shadow */
}

/* ===== Content Cards Section ===== */
.cards-container {
    display: flex;
    flex-wrap: wrap;
    justify-content: center;
    gap: 25px;
    padding: 40px 20px;
}

.card {
    background: linear-gradient(to bottom right, #ffffff, #dbe6ff); /* Gradient background */
    width: 280px;
    border: 1px solid #cccccc; /* Border */
    border-radius: 15px; /* Rounded Corners */
    margin: 10px; /* Margin */
    padding: 20px; /* Padding */
    box-shadow: 3px 3px 10px rgba(0, 0, 0, 0.2); /* Box Shadow */
```

```css
  text-align: center;
  transition: transform 0.3s ease, box-shadow 0.3s ease;
}

.card:hover {
  transform: translateY(-8px);
  box-shadow: 6px 6px 18px rgba(0, 0, 0, 0.35);
}

.card h2 {
  font-family: 'Courier New', monospace;
  color: #182848;
  text-shadow: 1px 1px 2px rgba(0,0,0,0.2); /* Text Shadow */
}

.card p {
  font-family: 'Arial', sans-serif;
  font-size: 14px;
  color: #444444;
}

.card button {
  background-color: #4b6cb7;
  color: #ffffff;
  border: none;
  padding: 10px 20px;
  border-radius: 8px; /* Rounded Corners */
  font-size: 14px;
  cursor: pointer;
  margin-top: 10px;
  box-shadow: 2px 2px 6px rgba(0,0,0,0.3);
  transition: background-color 0.3s ease, transform 0.2s ease;
}

/* Hover effect for buttons */
.card button:hover {
  background-color: #ffe600;
  color: #182848;
  transform: scale(1.08);
}

/* ===== Footer ===== */
footer {
  background-color: #1f1f2e;
```

```
    color: #ffffff;
    text-align: center;
    padding: 20px;
    margin-top: 30px;
    font-family: 'Verdana', sans-serif;
    font-size: 14px;
    text-shadow: 1px 1px 2px rgba(0,0,0,0.5);
}
```

7. CODE EXPLANATION

Code Element / Style Rule	Explanation
<canvas id="myCanvas">	Defines the graphic drawing surface region in HTML5.
getContext('2d')	Retrieves the 2D rendering context enabling drawing APIs (fillRect, arc, stroke).
ctx.fillRect / ctx.arc	Renders filled geometric structures (blue rectangle and yellow circle).
ctx.fillText()	Draws textual graphics natively onto the canvas surface with custom typography.
linear-gradient()	Applies smooth multi-color transitions to element headers and card backgrounds.
box-shadow / text-shadow	Provides visual depth, elevation, and realistic dropshadow effects to components.
border-radius	Rounds standard element edges to produce clean modern UI cards and buttons.
transition & transform	Adds smooth, high-performance visual state transitions on mouse hover (translateY, scale).
display: flex	Organizes individual content cards in a responsive multi-column dynamic row layout.

8. TEST CASES AND EXECUTION DATA

Test Case	Input / Action	Expected Result	Status
TC-01	Render HTML5 Canvas Element	Canvas renders blue rectangle, yellow circle,	Pass

		stroke line, and text 'Vijay Anudeep' cleanly.	
TC-02	Hover over Nav links	Background transforms to light blue (#4b6cb7) and text color shifts to yellow (#ffe600) with transition.	Pass
TC-03	Hover over Content Cards	Card shifts upward smoothly via translateY(-8px) and increases box shadow elevation.	Pass
TC-04	Hover over Card Buttons	Button background turns yellow (#ffe600), text shifts to dark navy, and scales up by 8% (scale(1.08)).	Pass

9. OUTPUT

10. RESULT

Thus, an interactive static webpage utilizing HTML5 Canvas graphic rendering and modern CSS3 visual layout styling (gradients, box shadows, transitions, and hover effects) was successfully designed and verified.